



Research on Academic Discourse Power of Scientific Collaboration on COVID-19 Pandemic

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ABSTRACT

The sudden outbreak of COVID-19 pandemic at the beginning of 2020 poses a significant threat to the health and safety of people worldwide. Given the speed and scope of the COVID-19 pandemic, countries around the world have carried out scientific collaboration in the fight against the COVID-19 pandemic. This paper divides academic discourse power into academic discourse right and academic discourse impact. The number of published papers reflects the discourse right, and the number of cites reflects the academic discourse impact. The visualization analysis of research papers from 2019 to 2020 describes the worldwide scientific collaboration on COVID-19, and the academic discourse power of authors, institutions, and countries can be studied from the perspective of scientific collaboration. We analyze the scientific collaboration of 27,851 papers related to COVID-19 published during 2019-2020 from the perspectives of authors, institutions, and countries by using HistCite and VosViewer. Pearson correlation analysis is used to study the correlation between scientific collaboration, the number of published papers, and the number of cites. Furthermore, we find that scientific collaboration positively correlates with academic discourse right and academic discourse impact. Based on analysis of author collaboration, institutional collaboration and country collaboration, it was concluded that China has the highest total cites, reflecting its high academic discourse impact during the COVID-19 pandemic, and the USA has the highest number of international collaborators and the highest total number of published papers, reflecting its high discourse right during the COVID-19 pandemic. The number of cites and the number of published papers are significantly positively correlated with the number of collaborators in Pearson correlation of author collaboration, institutional collaboration and country collaboration. This study has presented the global collaboration on the research of COVID-19. We compared academic discourse right and academic discourse impact across different levels of authors, institutions, and countries, concluding

that academic discourse right and academic discourse impact are significantly positively correlated with the number of collaborators.

CCS CONCEPTS

• **Networks**; • **Collaborative and social computing**; • **Human-centered computing**; • **Empirical studies in collaborative and social computing**;

KEYWORDS

Keyword: Academic discourse power, Academic discourse right, Academic discourse impact, Scientific collaboration
Additional Keywords and Phrases: Covid-19, Academic Discourse Power, Scientific Collaboration

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1 INTRODUCTION

The trend of globalization has become a basic feature of the current era. With international affairs being closely linked and the high population mobility, public health security becomes a hidden danger that cannot be ignored. Severe Acute Respiratory Syndrome (SARS) in 2003 was the first global public health emergency of the 21st century. The sudden outbreak of COVID-2019 at the beginning of 2020 poses a major threat to the health and safety of people in China and around the world, and has had a huge impact on all sectors of society. On January 30, 2020, in view of the worldwide impact of novel Coronavirus pneumonia (COVID-19), WHO declared the COVID-19 as an "international public health emergency". On January 30, 2020, the World Health Organization (WHO) declared the Novel Coronavirus pneumonia (2019-nCoV) outbreak an "international public health emergency" in view of the worldwide impact of the COVID-19 outbreak. With the spread of COVID-19 around the world, it has attracted great attention from scholars and experts in the field of scientific research and health at home and abroad. Relevant scholars began to study the relevant fields at the early stage of the pandemic. As of January 30, 2021, 27,851 papers related to the topic had been included in the WOS core.

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2 ACADEMIC DISCOURSE POWER

After the end of the "Cold War" in the 1990s, international competition began to shift from military strength to soft power such as economy and culture. International discourse power, as an important symbol to measure a country's soft power, has become a focus of academic attention and research in all countries [1]. As for the research of international discourse power, there are differences in the starting time and direction at home and abroad. Foreign research on international discourse power started relatively early, and separated from international political research, closely related to the concept of power, and a wide range of research fields. Marx and Engels first put forward the concept of "ideological discourse power" [2], believing that the ruling class obtains the political rights of the dominant position and realizes the discourse power of culture and ideology through dominating and mastering the whole social means of production and economic basis. Since the 1970s, "discourse" has been developing as an independent discipline. French philosopher Michel Foucault, for the first time, combined the linguistic concept of "discourse" with the political concept of "power" and proposed that "discourse is power" [3]. Discourse power refers to the right to speak, the right to speak, and the right to control public opinion [4]. This paper considers the academic discourse power consists of two parts: right and impact. Discourse right refers to the right to speak, which specifically refers to the qualification of scholars or academic groups to express opinions or thoughts in a certain research field, and the right of expression, participation and information in the research of the evaluation scientific application system of discourse power in China [5]. Discourse Impact refers to the effect of speech and thought, specifically refers to the power of scholars or academic groups to lead and control the speech in a certain research field. Academic discourse power requires scholars or academic groups to spread their academic thoughts to the audience in a certain carrier and in a certain form in a certain space and play a role to be recognized by the audience. Academic achievements are the carrier of academic discourse power, including academic monographs, academic papers, dissertations, academic reports, invention patents, etc. Since academic papers are the primary choice for scholars or academic groups to disseminate important findings or the latest research results, this paper chooses academic papers as the representatives of academic results to measure the academic discourse power of authors, institutions and countries in the medical field during the pandemic period. Discourse right is reflected by number of published papers and discourse impact is reflected by number of cites.

3 METHODS

3.1 Scientific Collaboration Networks

Scientific collaboration is referred to as one of the defining features of "Big Science" and one of the results of the "professionalization of science" [1]. Scientific collaboration networks of authors are considered connected if they have co-authored a paper. The coauthors' institutions and countries also connect due to authors' collaboration [2]. Scientific collaboration networks of journals can be determined based on citing-cited relationships between publications. The citing-cited relationships connect two journals [3]. The benefits and merits of research collaboration include: sharing and

transferring knowledge and research equipment, connecting scholars to a large scientific network, expediting the research process, and increasing the visibility of papers [4-9].

As collaboration has continued to increase, scholars have examined not only whether or not people are writing papers together, but if those individuals are employed by different institutions [10]. The development of modern science is characterized by significant differentiation and integration, which forces researchers to gradually abandon the original mode of individual research during the process of scientific research. To understand the situation of international collaboration in scientific research under the "international public health emergency", this study focuses on the current outbreak of COVID-19, taking WOS core periodical database as the data source, using VOSviewer and Hiscite to conduct a bibliometric analysis of papers with the theme of COVID-19 published in the year after the outbreak of COVID-19 pandemic. We constructed collaboration graphs for authors, institutions and countries in the medical field. Through the analysis of the international scientific research collaboration during the pandemic, we provided a global description of international scientific research collaboration when the world meets the COVID-19, an "international public health emergency".

3.2 Date source

The study analyzed 27851 documents from Web of Science (WoS) core collection. The type of the document is paper. Since the outbreak of the global COVID-19 public health emergency, it has rapidly attracted extensive attention from scholars all over the world. Since the World Health Organization and relevant departments in China did not adopt a unified standard name for "COVID-19" at the early stage of the pandemic, the academia did not form a unified name for it. On this basis, in order to make this study more convincing and ensure the rigorouslyness of scientific research, we selected the top 7 keywords from the list of keywords recommended by the Novel Coronavirus study related resources in the library of Wuhan University. The retrieval model is TS=) "COVID-19" OR "2019-nCoV" OR "nCoV-2019" OR "2019 Coronaviruses" OR "2019 novel Coronavirus" OR "Novel coronavirus" OR "Novel coronavirus pneumonia" ([11]. The time span was customized as 2019-2020, and the retrieval time was Jan 30, 2021. We filtered out papers which neither focus on the virus itself nor make medical contribution to combat viral infections but contained keywords related to COVID-19. 27851 papers were finally obtained.

4 AUTHORS' ACADEMIC DISCOURSE POWER IN COLLABORATION

Authors' collaboration or co-authorship networks document scientific collaboration through published papers, where nodes are authors and a link represent the fact that two authors have written at least one paper together. Co-authorship networks are thus undirected networks [12]. After the outbreak of novel coronavirus, scholars in the field of medical science quickly launched scientific research, and published the latest results of virus research through scientific research papers at the fastest speed. As can be seen from Figure 2, among the 27,851 papers obtained in this paper, 2909 were written by one author alone, accounting for 11% of the total. 3,439

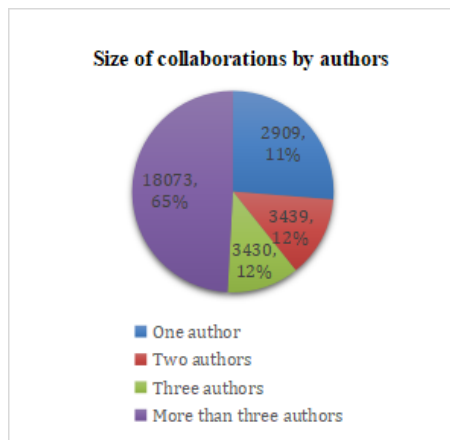


Figure 1: Size of collaborations by authors

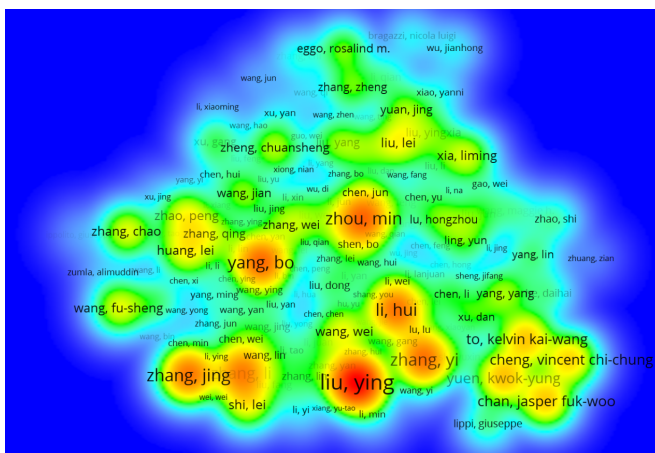


Figure 2: Authors collaboration density map

papers were co-authored by two authors, accounting for 12% of the total; 3,430 papers were co-authored by three authors, accounting for 12% of the total. There were 18,073 papers co-authored by three or more authors, accounting for 65% of the total. From the above analysis, it can be concluded that in the three months after the outbreak, most of the literature (74%) was produced by the collaboration of multiple authors. In order to display the current authors' collaboration for pandemic research, and then find the important author in the field of pandemic research group, we used VOSviewer to make a visualized analysis.

Authors collaboration network records scientific collaboration between authors through published papers. Nodes in the network are authors, and a link indicates that two authors have written at least one paper together. Authors collaboration network is undirected network [11]. After the outbreak of the pandemic, medical scholars quickly carried out a global scientific research on public health emergencies. In order to better reflect the current collaboration of domestic and foreign scholars on pandemic research and define the core group of scholars in the field of pandemic research,

the author imported the data from the WOS database into the visualization software VosViewer and selected the co-authorship for analysis. A map of collaborative knowledge among authors of relevant research papers was drawn one year after the outbreak of the pandemic (see Figure 3). A total of 129,822 authors (including multiple appearances by the same author) were included in 27,851 papers on COVID-19 in this study, and 184 authors with more than 10 papers were selected for visual analysis. The color of the node where the author is represents the density. The higher the density is, the darker the color is. The density depends on the density of the surrounding authors. The size of the author's name tag represents the number of cites. The more the number of cites, the greater the author's name tag is, the greater the discourse impact will be. By analyzing the density map of authors' collaboration, we can see the distribution of scholars with important research in the field of pandemic research. From the perspective of the density of collaboration among authors, within one year after the outbreak of the pandemic, Chinese scholars formed a large number of collaborative groups, and their research results were highly cited and had a high power of academic discourse worldwide.

In order to study the relationship between the number of collaborators, the number of cites and the number of papers, this paper makes use of SPSSAU to conduct Pearson correlation analysis on the number of papers published, the number of cites and the number of collaborators by 184 authors with more than 10 papers, and the specific results are shown in Table 1. The correlation analysis method was adopted to study the correlation among the number of cites, the number of papers and the number of collaborators. The Pearson correlation coefficient is used to indicate the strength of the correlation. The analysis shows that the correlation value between the number of papers and the number of collaborators is 0.461, and the significance level is 0.01, indicating that the number of papers and the number of collaborators is significantly positively correlated. The correlation value between the cites and the number of collaborators was 0.279, and the significance level was 0.01, indicating that the number of cites was significantly positively correlated with the number of collaborators.

5 INSTITUTIONAL ACADEMIC DISCOURSE POWER IN COLLABORATION

In this paper, a total of 26390 institutions were collected, and 380 institutions with more than 30 published papers and collaborative relationships were selected for research. The VOS clustering algorithm was used to analyze the correlation relationship and strength among the scientific research institutions. The research institutions were divided into 8 clusters, which reflected the collaborative groups of the research institutions under the international major public health emergencies. Clustering can be found from Figure 3, respectively is red group at Harvard medical school and the University of Washington as the core, Huazhong University of science and technology as the core and Wuhan university yellow group, and at the university of Oxford and the university of London as the core of green groups, with the core of the university of Toronto, shallow blue group, University of Milan as the core of the purple group. The dark blue group with the University of Melbourne and Monash as the core, the orange group with the University of Sao

Table 1: Correlation Analysis at Author’s Level

		Number of papers	Times cited
Number of collaborators	Pearson Correlation	.461**	.279**
	Sig. (2-tailed)	.000	.000
	N	184	184

** . Correlation is significant at the 0.01 level (2-tailed).

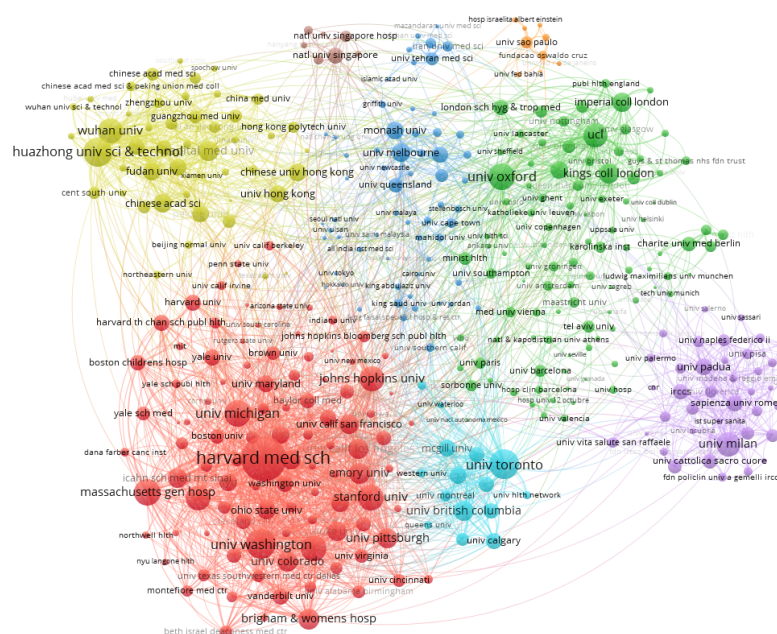


Figure 3: Institutional collaborative map

Paulo as the core, and the brown group with the National University of Singapore as the core.

In this paper, 380 institutions with more than 30 publications were selected for collaboration analysis. The number of papers published by scientific research institutions and their ranking are shown in Table 2, and the number of cites and their ranking are shown in Table 3. Among the top 10 institutions, 5 are from USA, 2 are from China, 2 are from the United Kingdom, 1 is from Canada, and 1 is from France. Scientific research institutions in these 5 countries have greater rights to speak on the research of COVID-19. The top ten scientific research institutions ranked by the number of cites are all Chinese, which reflects the high discourse impact of Chinese scientific research institutions in the research of COVID-19. It is worth mentioning that Huazhong University of Science and Technology ranks first in terms of the number of papers published and cites in the world, while Wuhan University ranks third in terms of the number of papers published and cites in the world which reflect their high discourse impact. According to the Table 4, correlation analysis shows the correlation value between the

number of papers and number of collaborators is 0.872 and shows a significant level of 0.01, indicating that the number of papers and number of collaborators have a significant positive correlation at the country's level. The correlation value between times cited and the number of collaborators is 0.430 and shows a significant level of 0.01, indicating that times cited and the number of collaborators had a significant positive correlation at the country's level. From the analysis above, it can be concluded that countries with many collaborators may also get a more significant number of papers and times cited.

6 COUNTRIES' ACADEMIC DISCOURSE POWER IN COLLABORATION

Research collaboration between countries can be reflected in the papers co-authored by researchers from different countries. This paper analyzes the collaboration network at the national level of all authors in 27851 scientific research papers (as shown in Figure 4). The size of the country node in the figure reflects the total cites

Table 2: Rank of papers published by scientific research institutions

Rank	Name of organization	Country	Number of papers	Cites	Number of partners
1	Huazhong Univ Sci & Technol	China	586	31273	631
2	Harvard Med Sch	U.S.A	457	6762	1273
3	Wuhan Univ	China	383	27989	484
	Univ Toronto	Canada	278	2287	669
4	Univ Milan	France	261	2709	460
5	Univ Oxford	Britain	261	6283	551
6	Columbia Univ	U.S.A	236	3743	507
7	Univ Washington	U.S.A	224	5318	653
8	Ucl	Britain	210	1999	466
9	Johns Hopkins Univ	U.S.A	199	2615	500
10	univ michigan	U.S.A	198	2188	554

Table 3: Institution list ranking by the number of papers published

Rank	Name of organization	Country	Number of papers	Cites	Number of partners
1	Huazhong Univ Sci & Technol	China	586	31273	631
2	Wuhan Univ	China	383	27989	484
3	Capital Med Univ	China	147	26631	357
4	Chinese Acad Med Sci	China	76	19528	171
5	Tsinghua Univ	China	86	19484	170
6	Chinese Acad Sci	China	168	17222	310
7	Univ Hong Kong	China	189	16009	277
8	Chinese Acad Med Sci & Peking Union Med Coll	China	54	11067	117
9	Shanghai Jiao Tong Univ	China	159	9862	246
10	Wuhan Jinyintan Hosp	China	30	6897	58

Table 4: Correlation Analysis at Institutional Level

		Number of papers	Times cited
Number of collaborators	Pearson Correlation	.872**	.430**
	Sig. (2-tailed)	.000	.000
	N	380	380

** . Correlation is significant at the 0.01 level (2-tailed).

of countries' published papers, and the line between countries reflects the collaboration relationship between researchers in the two countries. The thicker the line, the more frequent the collaboration. In this paper, 140 countries with more than 5 publications were selected for collaborative analysis.

As can be seen from the network diagram, China and USA have a larger node area, which proves that papers from these two countries have the largest number of cites, reflecting their high discourse impact. Moreover, countries are ranked according to the number of published papers in Table 4 which reflect the discourse right of countries. USA published the largest number of paper, reflecting its highest discourse right. And followed by China, Italy, England and other countries. According to the Table 6, correlation analysis shows the correlation value between the number of papers and number of collaborators is 0.894 and shows a significant level of

0.01, indicating that the number of papers and number of collaborators have a significant positive correlation at the country's level. The correlation value between times cited and the number of collaborators is 0.685 and shows a significant level of 0.01, indicating that times cited and the number of collaborators had a significant positive correlation at the country's level. From the analysis above, it can be concluded that countries with many collaborators may also get a more significant number of papers and times cited.

7 CONCLUSIONS

Through the analysis of collaboration at different levels, this paper mainly draws the following conclusions :(1) According to the analysis of author collaboration, Chinese scholars published the most papers and were the most cited in the one year after the outbreak

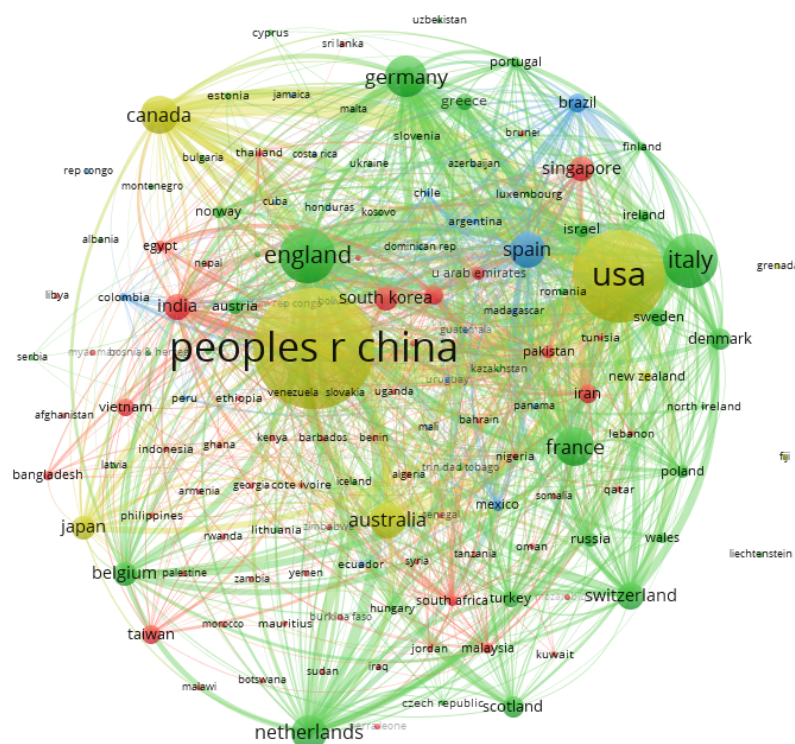


Figure 4: Map of research collaboration

Table 5: Countries list ranking by number of papers

Ank	Country	Papers	Cites	Number of International collaboration
1	USA	7660	75873	6306
2	Peoples r China	4187	131322	2808
3	Italy	2730	26620	3802
4	England	2593	28092	4905
5	India	1586	6536	1589
6	Spain	1419	11306	2564
7	Germany	1302	15319	2734
8	Canada	1244	13128	2397
9	Australia	1133	11458	2209
10	France	1028	13988	2251

Table 6: Correlation Analysis at Country's Level

		Number of papers	Times cited
Number of collaborators	Pearson Correlation	.872**	.430**
	Sig. (2-tailed)	.000	.000
	N	380	380

** . Correlation is significant at the 0.01 level (2-tailed).

of COVID-19. In terms of both discourse power and discourse impact, Chinese scholars took the lead in the research on COVID-19 in the world. The number of papers and the number of cites are significantly positively correlated with the number of collaborators. (2) According to the analysis of institutional collaboration, Chinese scientific research institutions were the most cited in the one year after the outbreak of COVID-19 pandemic, and the research on COVID-19 had the highest discourse impact. American scientific research institutions published the most papers, and had the most international collaboration frequency, so that they had the largest discourse right power. (3) From the analysis of collaboration by countries, we found the top ten countries in the world in terms of publication amount and total cites have all conducted scientific research collaboration with China and USA. USA has the largest publication amount so it has the largest power of academic discourse right. While China has the highest cites so it has the largest power of academic discourse impact. (4) According to Pearson correlation analysis, we find that scientific collaboration positively correlates with academic discourse right and academic discourse impact.

8 CITING RELATED WORK

This section cites a variety of journal [1–5, 7–11], book [6].

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